Analysis of removed genotypes (A. pisum dataset)

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Datasets

Libraries

```
library("tidyverse")
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.1     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library("patchwork")
```

Import of genetic cluster information

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
ivip_to_genetic_clusters <- read.csv(file = "../../../../00.info/02.genetic_structure_jonatha
                             header = TRUE, stringsAsFactors = FALSE, sep =",")
```

Import of A. pisum phenotyping screen data

Two data files:
- The first is the complete dataset *unfiltered*.
- The second is the filtered dataset, where filters have been applied:
* If no surviving adult was found, then observation was removed.
* If one adult was present but without any larvae, then the observation was removed.

Analysis was done for N>=6 biol. replicates (plants).

> **!** Important
>
> The complete data set contains 3027 rows and 249 genotypes.
> The filtered data set contains 2176 rows and 218 genotypes.

Comparison of proportions of genetic clusters in the raw and filtered dataset

List of IVIPs in original panel and raw/filtered A. pisum dataset

```
ivips_in_clusters <- unique(ivip_to_genetic_clusters$ivip)
ivips_in_raw_pisum_dataset <- unique(pisum_raw$ivip)
ivips_in_filtered_dataset <- unique(pisum_filtered$ivip)
```

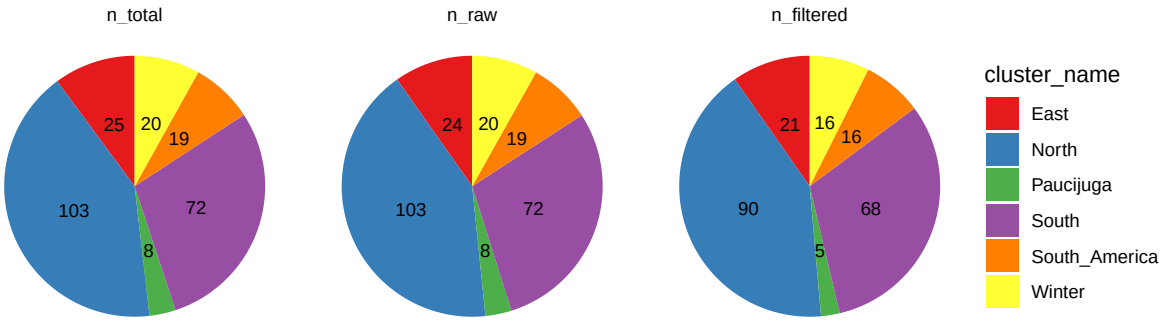
Data extraction

```
combined_counts_per_cluster <- inner_join(counts_per_genetic_cluster_total,
                                          counts_per_genetic_cluster_raw,
                                          by = "cluster_name") %>%
  inner_join(counts_per_genetic_cluster_filtered, by = "cluster_name") %>%
  # remove row if cluster_name == NAs
  filter(!is.na(cluster_name))


head(combined_counts_per_cluster, n = 10)
```

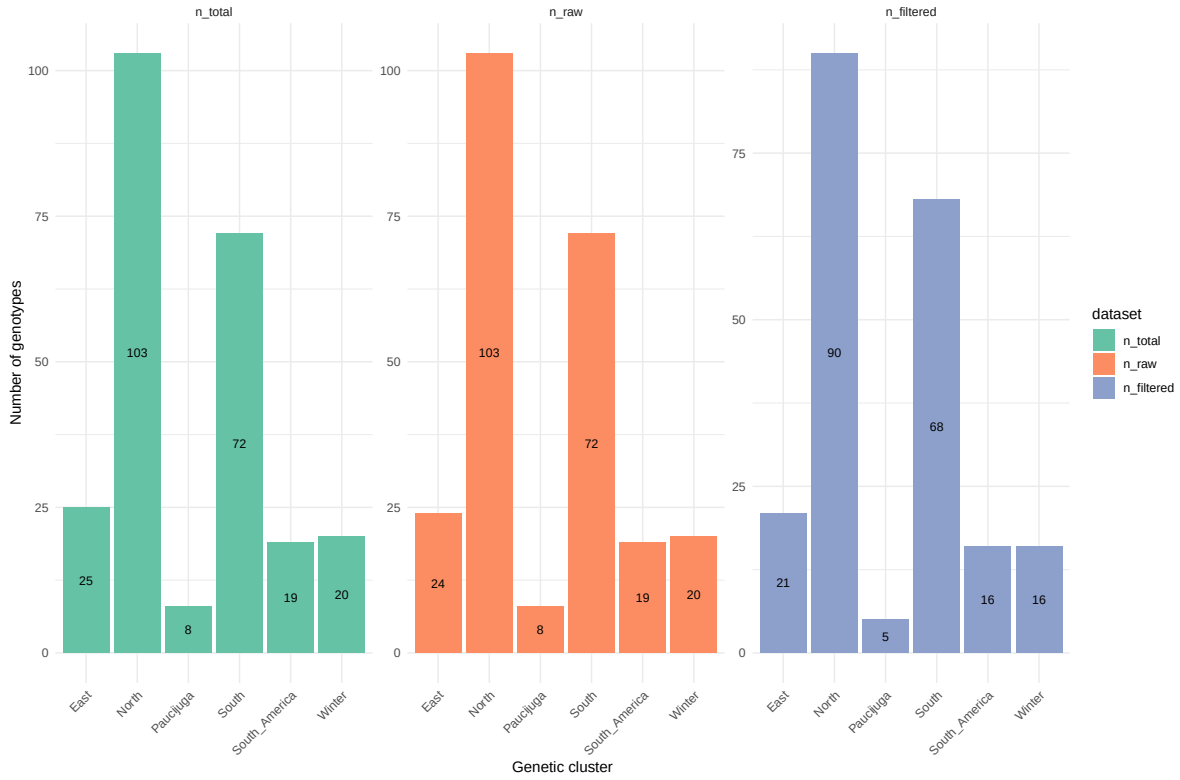
```
# A tibble: 6 x 4
  cluster_name  n_total n_raw n_filtered
  <chr>           <int> <int>      <int>
1 East               25    24         21
2 North             103   103         90
3 Paucijuga           8     8          5
4 South              72    72         68
5 South_America      19    19         16
6 Winter             20    20         16
```

Pie chart



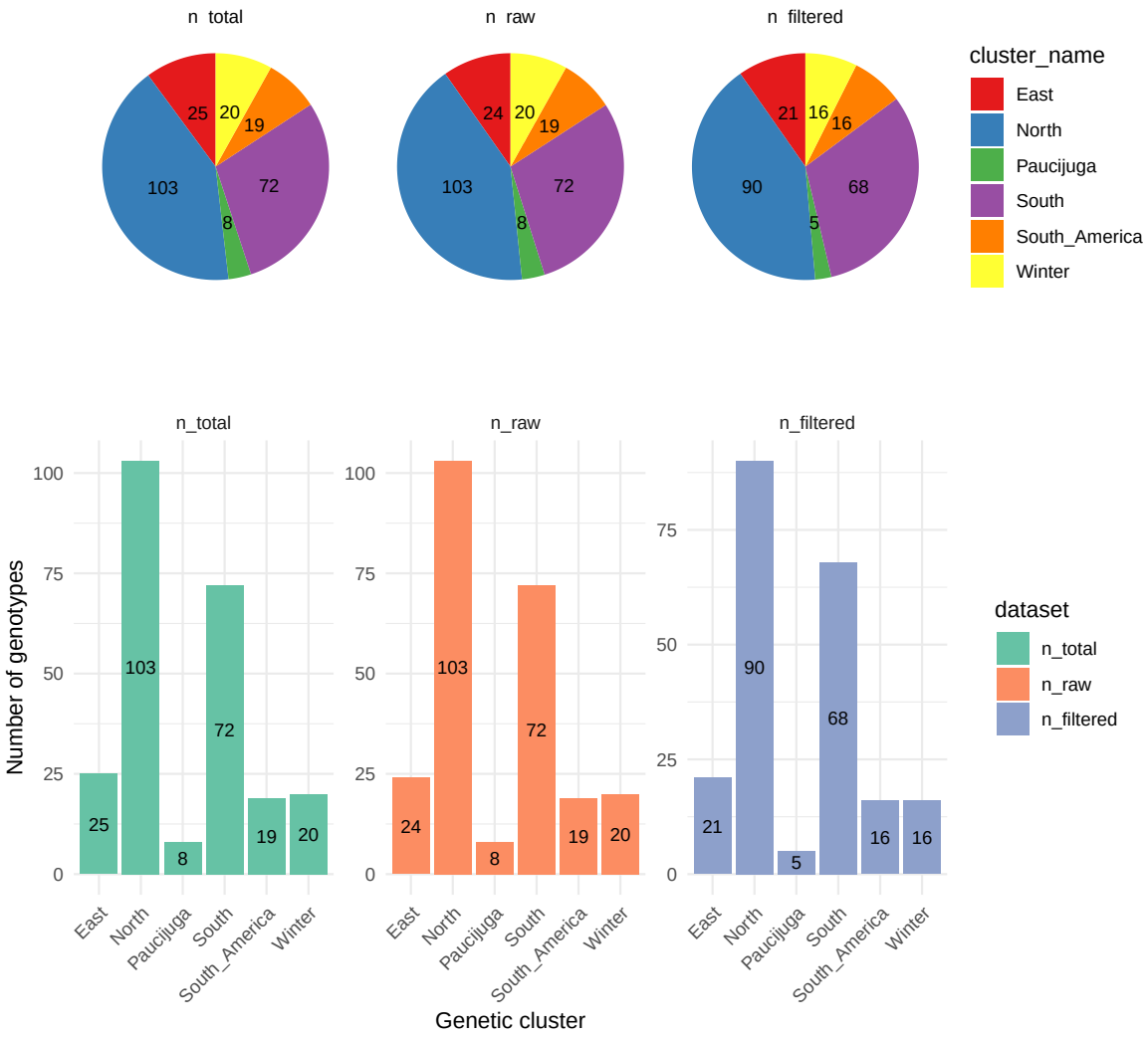
Barplot

Often a barplot is much easier to compare.



Combined plot (pie chart + barplots)

Distribution of genotypes in genetic clusters (A. pisum)



! Important

We see that the proportions are not different between the complete set of genotypes (N=248), the set of genotypes from which observations of aphid fecundity were made and the final filtered dataset.

Save plots.

```
ggsave("./acyrthosiphon_pisum_genetic_clusters_distribution.pdf",
       width = 8, height = 8)
ggsave("./acyrthosiphon_pisum_genetic_clusters_distribution.png",
       width = 8, height = 8, dpi = 300)
```